

Predicting Loan Default Risk

Using LendingClub Data

Course:	Data Mining
Type:	Binary Classification
Dataset:	LendingClub (2007 - 2018 Q4)
Records:	1,345,310 terminal loans
Features:	132 predictors (after preprocessing)

1 · Introduction & Problem Statement

The Problem

LendingClub assigns grades A-G to each loan, but grades alone don't capture true default probability at the individual loan level.

Can a machine learning model predict whether a loan will be Fully Paid or Charged Off — and beat LendingClub's own grading system?

Research Objectives

- Build a binary classifier (Fully Paid vs. Charged Off)
- Use only features available at loan origination
- Evaluate model lift vs. LendingClub's grade signal
- Identify key default drivers via SHAP explainability

Why It Matters

- Consumer lending involves substantial financial risk — misclassifying defaults is costly
- A better risk signal allows lenders to minimize losses while extending credit fairly
- Interpretable models build trust with regulators and borrowers alike

2 · Dataset Overview

1.68 GB

Raw file size

1 CSV, 151 columns

2,260,701

Total records

Accepted loans 2007-2018

1,345,310

After filtering

Fully Paid + Charged Off only

~20%

Default rate

Class imbalance present

Key Feature Groups (at origination):

Credit Profile:

FICO, inquiries (last 6 mo.), delinquencies, revolving utilisation, pub records

Loan Characteristics:

Loan amount, interest rate, term, grade, sub-grade, purpose

Borrower Profile:

Annual income, DTI, employment length, home ownership, state

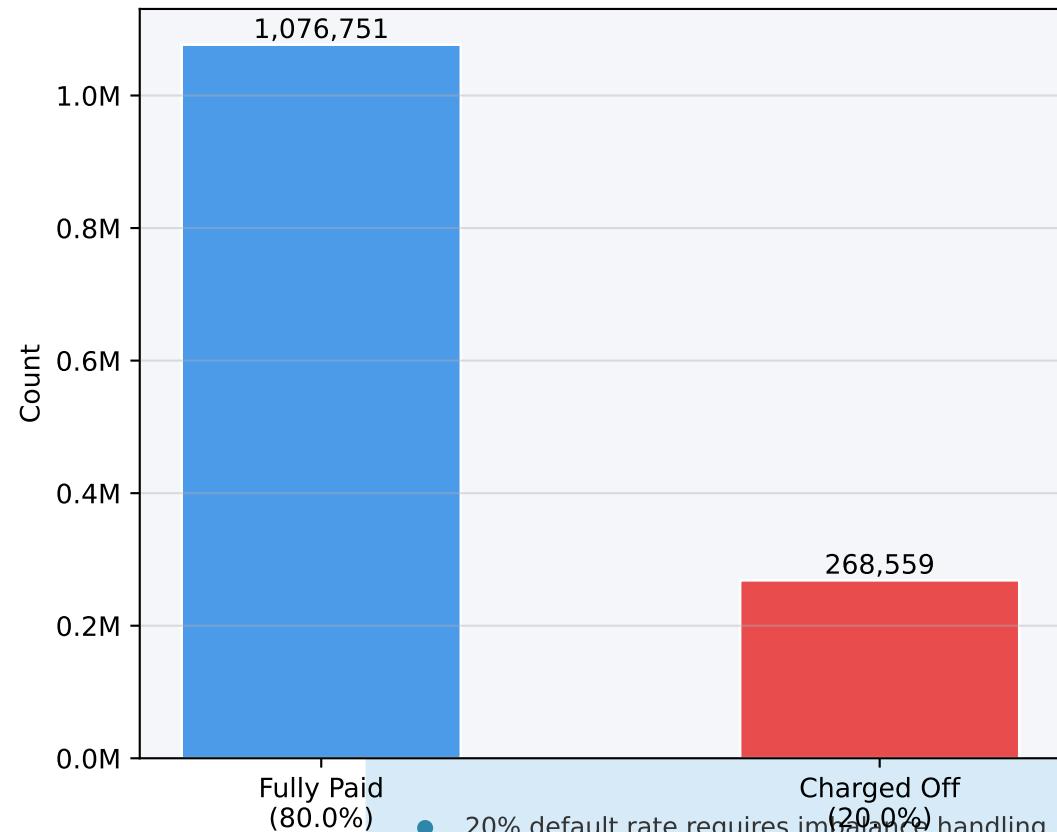
Engineered:

Loan-to-income ratio, credit age (months), FICO midpoint average

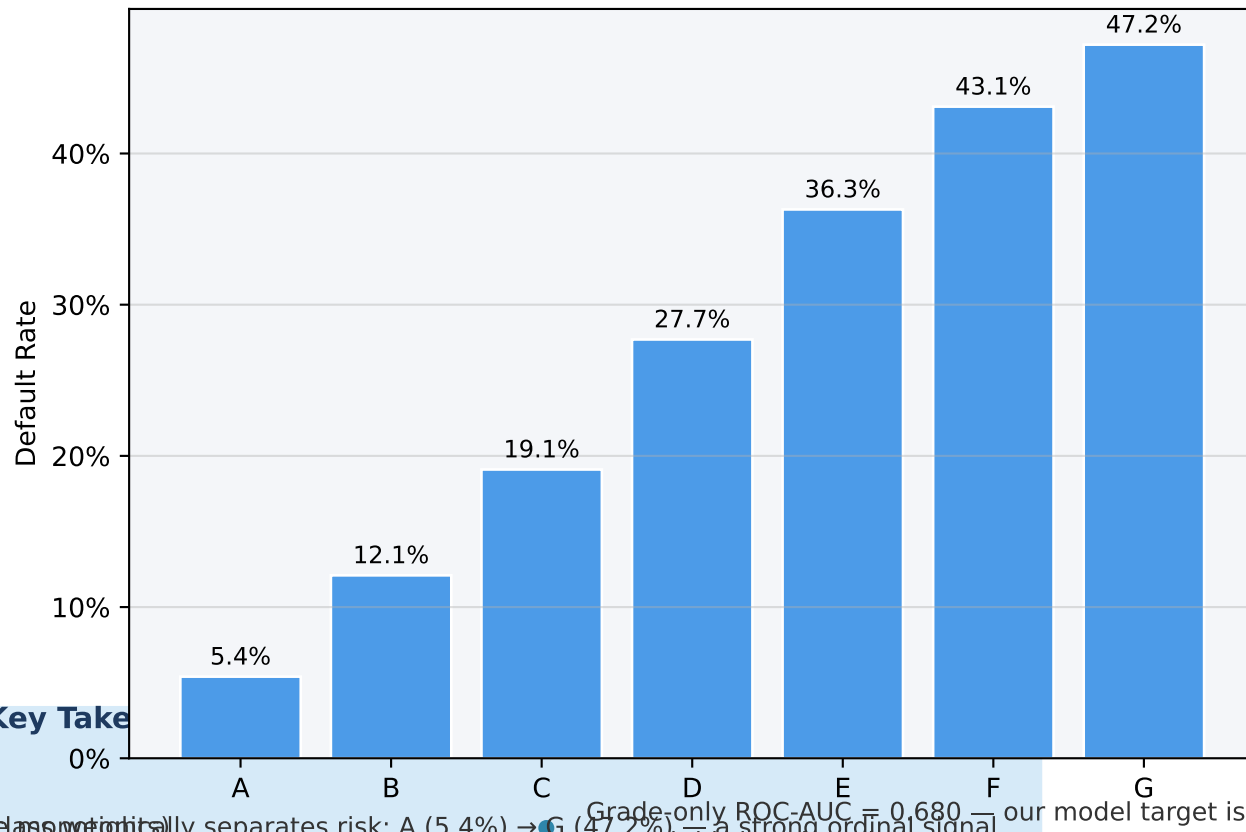
Post-origination columns (recoveries, total_pymnt, out_prncp, etc.) excluded to prevent data leakage

3 · Exploratory Data Analysis — Class & Grade

Class Distribution (n = 1,345,310)



Default Rate by LendingClub Grade



Key Take

● 20% default rate requires imbalance handling (SMOTE, class weights) — Grade-only ROC-AUC = 0.680 — our model target is to exceed this benchmark

3 · Exploratory Data Analysis — Feature Insights

Top Correlations with Default (target):

int_rate	+0.31	Higher interest = higher risk
sub_grade	+0.30	Grade encoding captures risk ordering
grade	+0.28	Coarse version of sub_grade
dti	+0.11	Higher leverage → more defaults
fico_range_low	-0.13	Lower FICO → more defaults
annual_inc (log)	-0.09	Higher income → fewer defaults

Other Notable Findings:

- 60-month loans default at ~26% vs. ~16% for 36-month loans
- Small-business and moving loans show highest default rates by purpose
- Rejected applicants have higher DTI & lower FICO vs. accepted borrowers

73 of 151 columns had missing values; 35 columns dropped (>50% missing)

4 · Preprocessing Pipeline

1 Column Selection

Exclude 37 post-origination + 10 non-predictive columns at load time
151 raw columns → 102 loaded

2 Target Filtering

Keep only Fully Paid / Charged Off rows
2,260,701 → 1,345,310 rows | default rate: 19.96%

3 Missing Value Drop

Drop 35 columns with >50% missing rate
(e.g. annual_inc_joint, dti_joint, mths_since_last_major_derog)

4 Feature Engineering

$\text{loan_to_income} = \text{loan_amnt} / \text{annual_inc}$
 $\text{credit_age_months} = \text{issue_d} - \text{earliest_cr_line}$
 $\text{fico_avg} = (\text{fico_range_low} + \text{fico_range_high}) / 2$

5 Encoding

Ordinal: grade (1-7), sub_grade (1-35), emp_length (0-10), term (36/60)
One-hot: home_ownership, purpose, verification_status, addr_state

6 Imputation

Median imputation for 42 remaining numeric columns with NaN
→ 0 nulls remaining

Final dataset: 1,345,310 rows × 133 columns | 0 nulls | all numeric

5 · Modeling Approach

Train / Test Split

Stratified 80/20 split → Train: 1,076,248 rows | Test: 269,062 rows

Both sets maintain 19.96% default rate

Three Models Trained:

Logistic Regression (Baseline)

Parameters:
class_weight='balanced'
SAGA solver, C=0.1
StandardScaler pipeline

Why:
Simple, interpretable baseline
Handles imbalance via weights

Random Forest

Parameters:
500 trees, max_depth=12
min_samples_leaf=50
class_weight='balanced_subsample'

Why:
Non-linear, robust to outliers
Imbalance via per-tree resampling

XGBoost (Best Model)

Parameters:
500 estimators, lr=0.05
max_depth=6, subsample=0.8
scale_pos_weight=4.01

Why:
Gradient boosting with
native imbalance correction

Best Performer

Imbalance Strategy Comparison

SMOTE (synthetic over-sampling) was tested against scale_pos_weight.
SMOTE ROC-AUC = 0.7314 vs. XGBoost native = 0.7342 — native weighting preferred.

6 · Model Results

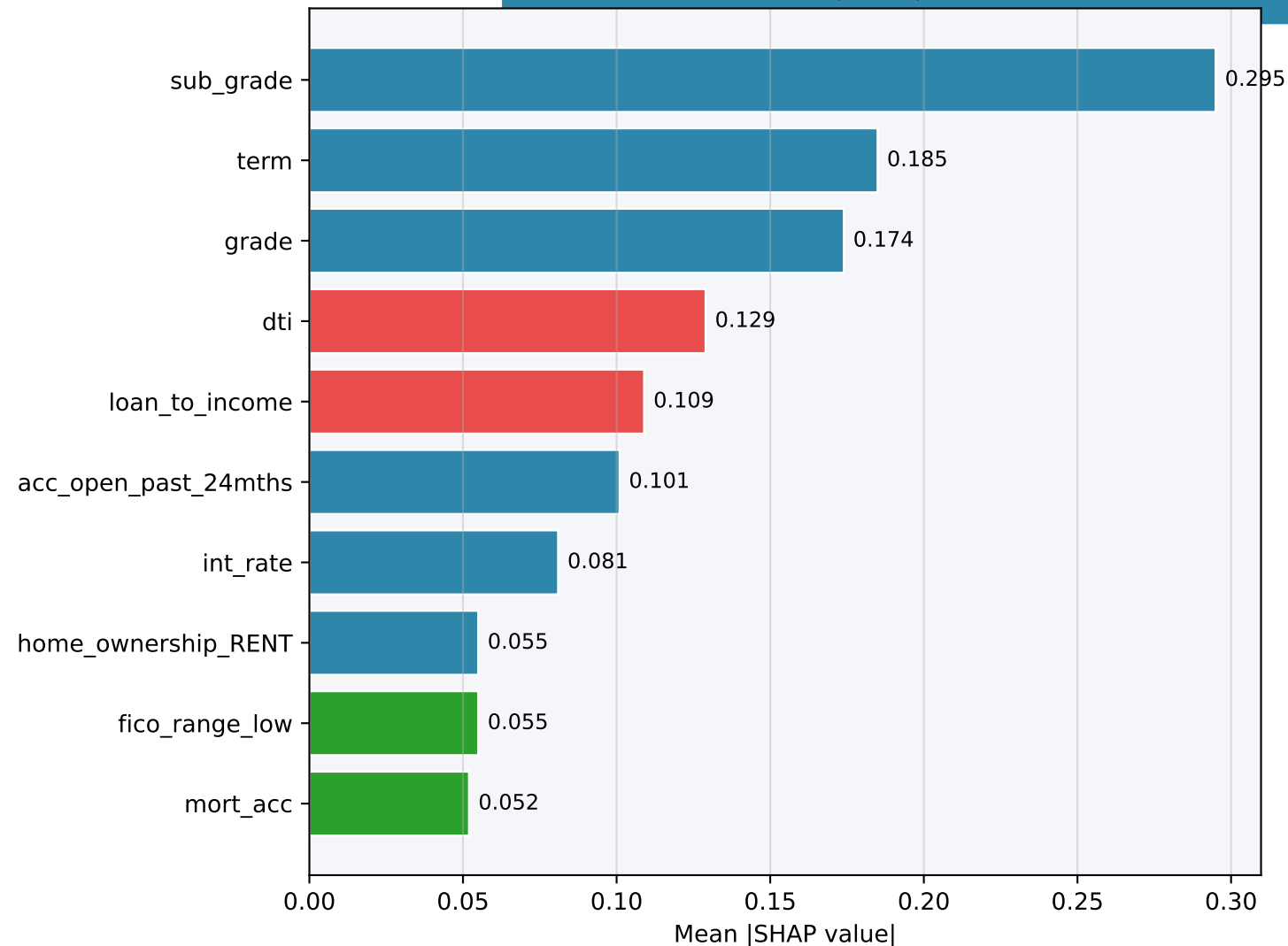
Model	ROC-AUC	PR-AUC	F1 (t=0.5)
Logistic Regression (balanced)	0.7186	0.3828	0.4359
Random Forest	0.7175	0.3859	0.4326
XGBoost	0.7342	0.4095	0.4461
XGBoost (optimal threshold)	0.7342	0.4095	0.4485
Grade-only baseline	0.6801	—	—

5-Fold Cross-Validation (ROC-AUC):

Logistic Regression 0.7178 ± 0.0014	Random Forest 0.7172 ± 0.0013	XGBoost 0.7333 ± 0.0009
---	---	-----------------------------------

7 · SHAP Explainability — What Drives Defaults?

Top 10 Features — Mean |SHAP| (XGBoost)



Feature Interpretations

sub_grade / grade / int_rate

LendingClub's own risk signal is the strongest predictor — but the model adds information beyond it

term

60-month loans carry significantly higher default risk — second most influential feature

dti + loan_to_income

Borrower leverage at origination is a key risk driver; our engineered ratio adds signal beyond raw DTI

acc_open_past_24mths

Recent credit-seeking behaviour signals financial stress

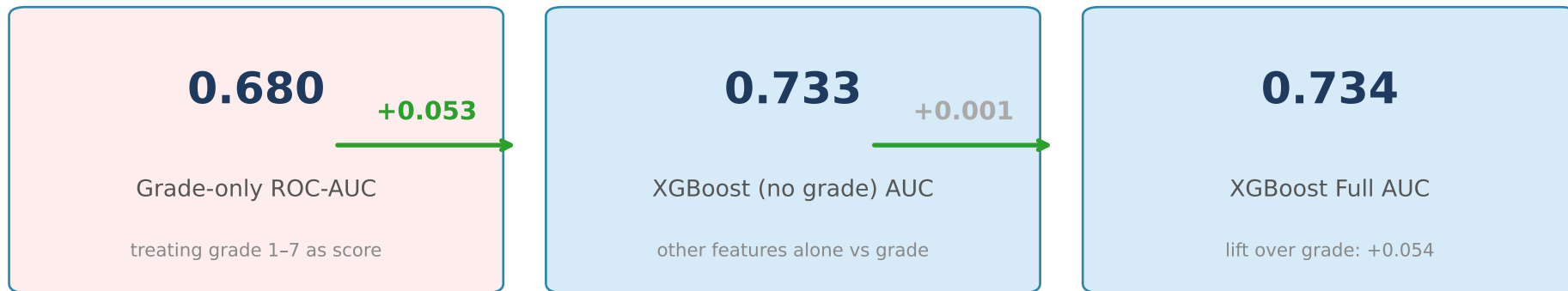
fico_range_low

Lower FICO → higher default; negative SHAP direction (as expected — strong inverse relationship)

mort_acc

More mortgages → lower default risk (wealthier borrowers)

8 · Model vs. LendingClub Grade



What This Tells Us

- The model provides a meaningful +5.4% AUC lift over LendingClub's grades alone
- Even without grade/sub_grade, other borrower features achieve 73.3% AUC — beating grade-only by +5.3%
- Adding grade back improves AUC only marginally (+0.1%) — most grade signal is already captured by correlated features
- Conclusion: a data-driven model significantly outperforms the existing grade system as a default risk signal

9 · Future Work & Limitations

Potential Next Steps

Hyperparameter Tuning

Bayesian search over XGBoost parameters (max_depth, min_child_weight, gamma, learning_rate) — expected additional AUC improvement of 0.01-0.02

Neural Networks / TabNet

Deep tabular models may capture higher-order feature interactions that tree models miss in the 132-feature space

Calibration

Platt scaling or isotonic regression to produce well-calibrated probabilities — currently AUC is strong but probability values need calibration check

Business Threshold Optimisation

Tie decision threshold to actual financial cost (loss given default, LGD) rather than maximising F1 alone — different cost for FP vs. FN

Temporal Validation

Validate on a strict time-based split (train 2007-2015, test 2016-2018) — prevents any future data leaking through time

NLP on Loan Purpose Text

The 'desc' / 'title' free-text columns were dropped — NLP embeddings could extract additional signal

Key Limitations

- ▲ Data covers 2007-2018 — economic conditions may have changed since then
- ▲ Rejected applications have fewer features — full selection-bias analysis not possible
- ▲ SMOTE on 1M+ rows is computationally expensive — class weights used as practical alternative

Summary & Conclusions

Dataset

1.34M terminal loans, 132 features, 20% default rate

EDA

Grade, interest rate, DTI, and FICO are the dominant risk signals

Pipeline

Post-origination leakage prevented; 3 engineered features added

Modeling

XGBoost (ROC-AUC 0.734) outperforms LR and Random Forest

vs. Grade

Model lifts AUC by +5.4% over LendingClub's grade-only baseline

SHAP

sub_grade, term, dti, and loan_to_income are top model drivers

A machine learning approach to credit default prediction adds measurable value beyond traditional rule-based grading systems — with full interpretability via SHAP.